

Introduction to R

Project management in RStudio

- RStudio is an integrated development environment (IDE).
- R is integrated into RStudio.

RStudio provides a useful feature: creating a self-contained working space (i.e. working directory).

You can create projects in existing directories (folders) or create a new one.

Creating RStudio Project

To create a project, go to:

- File
- New Project
- New directory
- Name your project : data-carpentry
- Create project

Organising working directory

- `data/` - your raw data. **READ ONLY**
- `data_output/` - your data output **READ AND WRITE**
- `documents/` - the documentation associated with the project (e.g. cookbook)
- `fig_output/` - your figure outputs **WRITE ONLY**
- `scripts/` - your code **READ AND WRITE**

Two main ways to interact with R

- Test and play environment within the interactive **R console**
- Write and save an **R script (.R file)**

	Console	R script
Pros	Immediate results	Work lost once you close RStudio
Cons	Complete record of your work	Messy if you just want to print things out

Vector

- Store multiple values in one object
- All values have to be of the same data-type
- `c(100, 200, 300)`
- `c(Apples, Pears, Oranges)`
- **Note:** vector data in the geospatial context is different from vector data!

Data types

Data type refers to a type of information that is stored by a value. It can be:

- `numerical` (a number)
- `integer` (a number without information about decimal points)
- `logical` (a boolean - are values TRUE or FALSE?)
- `character` (a text/ string of characters)
- `complex` (a complex number)
- `raw` (raw bytes)

Data structures

Besides vectors you will also come across:

- **Data frames:** two-dimensional data structure | combination of vectors of the same length
- **List:** can have arbitrary data structure and can mix with data types
- **Matrices:** two-dimensional data structures containing elements of the same data type

Exercise

Combine the `abcd_vector` with the `numeric_vector`

What is the data type of this new vector, and why?